

Connecting Small Worlds with CSF and Neural Plasticity Modulators: A Mini Proof of Concept in Rats

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What Is the Idea?

Scientists found that a natural substance in the fluid around young brains, called FGF17, seems to help old mice remember things better. Another substance, called 7,8-DHF, is thought to help brain cells make stronger connections with each other. We want to know: if we give both, and also give the animals a rich, stimulating place to learn, do they become better at solving different kinds of problems than normal rats of the same age?

What Will We Do?

This is a deliberately small version. We use 30 healthy young rats in three groups of 10: normal rats; rats that spend time in the learning environment; and rats that get the two substances and the learning environment. Another 22 rats are helpers (12 to check that FGF17 really does something in the rat brain, 6 for practice and safety checks, and 4 for trial runs of the tasks), and up to 12 more are held in reserve for one repeat of that check. The 30 main rats do the same three tasks (we expect to lose some along the way, so only about 18 to 21 may finish) — a puzzle where the rules suddenly change, a maze with new problems, and a swimming-pool memory test. Two halves of the group take the tasks in a different order at the same time, to save weeks. The check on FGF17 comes first: if there is no sign that it works, we stop and save the money. The rats get a small operation that places a thin tube into the brain's fluid space, daily injections for five weeks, limited food for some weeks and swimming tests, and are humanely put to sleep at the end; an animal-ethics committee must approve everything first.

Who Does the Work?

A suitable host site with a licensed animal facility (for example a university, clinic, hospital research unit or contract research organisation) runs the experiment, following this document and detailed lab procedures that the researcher approves. The researcher guides it from a distance and keeps the secret group codes, so the staff giving the medicines do not know which rat got what, and checks the results independently.

How Much Will It Cost?

Materials only (animals, chemicals, equipment parts, supplies) are planned at about USD 9,000 to 15,000 (the lower end if the host already owns the apparatus); import duties and fees vary by country and are not included, and the planned materials fit a budget of up to about USD 15,000. In total we plan 52 rats (up to 64) over about 8 to 9 weeks at the host site. Staff pay and the host institution's running-cost charges are NOT included. The real total might be about USD 25,000 to 65,000 (an unverified estimate); the host must quote it.

What Will This Test Tell Us — And Not Tell Us?

It can tell us whether FGF17 shows a chemical sign of working in the rat brain, whether the lab can run all the tasks smoothly, and whether the results lean in a promising direction. With so few rats only a very large effect would show. Comparing the third group with the second shows what the substances add on top of the environment, but the substances are not tested alone. It cannot prove the idea works or say anything about people.

What If It Goes Well?

Then we plan a bigger experiment: first separating what each ingredient does, then using optogenetics (a technique that switches chosen brain cells on and off with light) to see how brain activity interacts with these substances.

The Long-term Dream: Brain Longevity

The original finding was in old mice: young brain fluid helped their memory. With ageing, and in diseases such as Alzheimer's, the bridges between brain cells and their insulation wear away, fewer new cells may form, and the brain's small-world organisation and its rhythmic electrical pulses lose their order. "Small world" is a network idea (Watts and Strogatz, 1998): like a town of tight neighbourhoods linked by a few long roads, the brain has dense local clusters of cells that talk mostly to each other, plus some long-range links that let any region reach any other in only a few steps. Brain-imaging studies describe this pattern in human brains, and it is thought to let the brain do specialised and combined work at once, cheaply; it is reported to change in ageing and in Alzheimer's. 7,8-DHF is thought to help cells build and strengthen the connections that form these clusters and links; that is the reason for including it. The long-term hope is brain longevity: to test whether a treatment like this could help keep those bridges and their insulation, support new cells where the adult brain makes them, and preserve the brain's rhythm architecture into old age, so that thinking and memory decline less; and, in the same spirit, help people with learning difficulties or problems with how brain cells connect. That is a hope, not a result. This study uses healthy young rats and measures none of this (no connections, no new cells, no brain rhythms, no ageing, no disease). The next steps would be old rats and disease models with those measurements, then safety work, and only much later people; many promising brain treatments have failed on that road, and getting these substances into a human brain safely is a big obstacle. The rest of this document gives the technical details.

Technical protocol (mini proof of concept; core materials about USD 9–15k)

Does a candidate plasticity-promoting treatment (FGF17 + 7,8-DHF) combined with a structured learning environment raise performance across different cognitive tasks in rats? A minimal, staged proof-of-concept study run by a host site under remote oversight.

Researcher / sponsor	Solano Moreira Furlan (remote; the host site runs the experiment)
Document	Mini proof-of-concept protocol, version 6 (final after five review rounds). The task methods, statistics logic and many safeguards come from a larger design reviewed in eight rounds in an earlier form and one round in a full-size form; this mini design is reviewed in the rounds listed in section 11.
Date	9 October 2026
Status	Not approved by an ethics committee. Local regulatory, import and price details come from search snippets and reviewer knowledge and were not verified; the host must confirm them. Prices are estimates, not quotes. “crf” in the request was read as CSF, cerebrospinal fluid (assumption to confirm).

Executive Summary (Technical)

Question. Does a candidate plasticity-promoting treatment — FGF17 (identified in young cerebrospinal fluid; Iram 2022 [1]) plus 7,8-DHF (a TrkB/BDNF-pathway agonist [2–4]) — combined with a structured enriched learning environment (ELE) raise the performance of healthy young rats across three tasks, one of which (set-shifting) uses stimulus dimensions not practised in the ELE, compared with untreated peers?

Design. One wave. Three arms of 10 enrolled rats (30 main rats; 6–7 expected completers per arm): C (vehicle, standard housing), E (vehicle + ELE) and D+E (FGF17 + 7,8-DHF + ELE). There is no drug-only arm, so drug and environment cannot be separated; the study compares the combination with enrichment alone and with untreated peers. 52 rats in total (30 main, 12 FGF17-engagement satellites, 6 practice/tolerability, 4 unoperated pilot rats), plus 12 contingency rats registered as a maximum (64 at most; 56 cannulated). Phase 0a (reagent activity and engagement; a gate) precedes Phase 0b (behaviour). The two halves of the cohort (15 rats each, balanced by arm) run the three tasks in parallel and in different orders, which shortens the calendar to about 8 weeks. The study informs, and does not confirm, whether a larger study is justified.

Budget (honest). Materials-only core about USD 9–15k (the lower end if the host already owns the arenas and mazes and has favourable animal facility rates; the upper end in pessimistic cases). The planned materials fit a budget of up to about USD 15k. Import duties and fees vary by country and are not included. Personnel, service fees, institutional overhead, equipment fees and a currency reserve are not included and must be quoted by the host; the all-in cost is probably USD 25–65k (unverified).

1. Aim, Hypotheses and What the Data Can Claim

Primary aim (feasibility): show that the reagent is active and engages its pathway in vivo; show that a host site can run the ELE and all three tasks with an acceptable completion rate; estimate variance and the direction of the D+E – C contrast.

Exploratory hypothesis: D+E performs better than control peers (C) and than enrichment alone (E).

Transfer: set-shifting is the primary transfer readout (odour and digging-medium dimensions are excluded from the ELE). Hebb-Williams and the water maze are spatial; the ELE has tunnels, climbing and wheels, so they are flagged as spatial, not transfer.

Allowed claims: feasibility, attrition, variance, direction with a 95% interval of D+E – C and D+E – E (overall index; set-shifting reported separately), and pathway engagement (pERK at the ventricular interface) — for the drug package as a whole, in combination with the ELE.

Not allowed: “smarter”, “general intelligence” or a g factor [8,9]; neuroplasticity; single-agent claims; any claim about the drug alone (there is no drug-only arm; D+E – E estimates the drug package on top of enrichment); TrkB-specific claims (pERK shows pathway engagement only); human, ageing or disease claims; any claim about network organisation, brain rhythms or new neurons; replication of Iram 2022 (an aged-mouse rescue; this asks about enhancement in healthy young adults, which no study found has tested).

Abbreviations. C, E, D+E: control, enrichment-only and drug-plus-enrichment arms; CSF: cerebrospinal fluid; ELE: enriched learning environment; ICV: intracerebroventricular (into the brain’s fluid space); IP: intraperitoneal; pERK/ERK: phosphorylated and total extracellular-signal-regulated kinase; aCSF: artificial cerebrospinal fluid; BSA: bovine serum albumin; DMSO: dimethyl sulfoxide; CMC-Na: sodium carboxymethylcellulose; SD, CD, ID1, ID2, ED in set-shifting: simple and compound discrimination, first and second intradimensional shifts, extradimensional shift (SD elsewhere is standard deviation, g is Hedges’ g); OPC: oligodendrocyte precursor cell; OC: operating characteristics; CV: coefficient of variation of the assay; FX: foreign exchange; PK: pharmacokinetics; MCI/AD: mild cognitive impairment/Alzheimer’s disease. KR: Kenward-Roger degrees of freedom; ICC: intraclass correlation; IPTT: implanted subcutaneous temperature transponder; OSF: Open Science Framework (pre-registration); AAV: adeno-associated virus; N_cap: the ED trial cap. References are listed by topic, not by order of first citation.

2. Long-term Goal, Next Experiment (Including Optogenetics) and Translational Roadmap

Recommended wording (translational reviewer, adopted):

“Our long-term aim (brain longevity) is to test whether combining a putatively plasticity-promoting biological treatment (an FGF17-type factor plus a TrkB/BDNF-pathway agent whose mechanism is validated; 7,8-DHF currently is not) with structured enriched learning can produce measurable cognitive benefit in humans. We pursue two **separate** programmes with distinct biology, models, ages and ethics: (a) neurodevelopmental conditions with altered connectivity (learning disabilities), and (b) age-related cognitive decline and early Alzheimer’s disease. We hypothesise that supporting synaptic and myelin plasticity is **one possible contributor** to such benefit; this is a hypothesis to be demonstrated with structural and functional evidence, not an assumption, and it is not addressed by the present study. Later stages would also ask whether such treatment preserves long-range and local (small-world) network organisation, supports new-cell formation where it occurs in the adult brain (adult neurogenesis is itself debated, especially in humans, and the FGF17 evidence concerns oligodendrocyte-lineage cells and myelination, not neurons) and preserves oscillatory rhythms (for example hippocampal theta and gamma, measured by local field potentials or EEG) in ageing; none of these is measured here. Prior neurotrophin programmes in Alzheimer’s have not shown benefit (including a direct-delivery AAV-NGF trial), brain delivery of protein factors is a major obstacle, and both candidate agents face unresolved questions (human CNS delivery of FGF17; whether 7,8-DHF engages TrkB). Each stage proceeds only on pre-registered target-engagement, structural and safety gates.”

Stage	Content	Gate to proceed
0 – this study	0a engagement/reagents → 0b behaviour in healthy young rats (mini proof of concept; tissue banked, histology not included).	Section 8.
1a – next experiment: powered rat confirmation	Both sexes; full factorial separating FGF17, 7,8-DHF and enrichment (and exercise); sustained infusion by pump; dose-response; durable vs on-drug effects; TrkB and FGFR necessity tests; synaptic readouts (PSD-95/synaptophysin on banked tissue, LTP).	Replicated effect on a pre-registered primary; structural effect.
Stage	Content	Gate to proceed
1b – next experiment: optogenetics (only if 1a is positive; mechanistic, not translational)	Question: how does controlled neuronal activity interact with FGF17 and with 7,8-DHF-driven plasticity? Candidate designs (to be designed and reviewed in their own round): (a) <i>activity × plasticity-opening</i> — optogenetic stimulation of a defined circuit (e.g. CaMKIIa-driven channelrhodopsin in prefrontal or hippocampal pyramidal neurons) during or after learning, with and without FGF17/7,8-DHF, asking whether activity amplifies	1a positive; opsin tools validated in the host site.

Stage	Content	Gate to proceed
	oligodendrocyte-precursor proliferation, myelination and synaptic plasticity (mouse precedent: premotor-cortex stimulation, Gibson 2014 [10]; extension to prefrontal/hippocampal circuits is an extrapolation); (b) <i>necessity</i> — inhibitory opsins during learning or consolidation; (c) <i>cell-type targeting</i> of oligodendrocyte-lineage cells needs Cre lines that are mostly mouse (species switch or an AAV-based rat strategy). Requirements: AAV vectors with institutional biosafety approval (institutional biosafety committee and any national genetically-modified-organism licence required by local regulations; the facility's biosafety licence must cover genetically modified animals at the right containment level), fibre implants, calibrated light sources, opsin-negative/fluorophore-only and light-only controls (heating, visual artefacts), histological verification of opsin and fibre placement, and monitoring for stimulation-induced seizures/kindling. Beyond this budget; own funding, equipment, expertise and ethics approval.	
2 – disease models (separate tracks)	Aged rodents (Iram-type rescue); Alzheimer's models (5xFAD, APP/PS1, TgF344-AD) — amyloid-model rescue has repeatedly failed to translate; neurodevelopmental models at appropriate ages: Fmr1 knockout (fragile X) and Down-syndrome models (Ts65Dn is being superseded) as <i>separate tracks</i> .	Benefit with engagement and structural evidence.
3 – drug development	Clinically deliverable form (intranasal, intrathecal or engineered/receptor-selective agent — unproven); a validated TrkB-pathway drug with human-relevant PK; toxicology in two pharmacologically relevant species where they exist (one if only one is relevant) for a protein biologic under ICH S6(R1), and per ICH M3(R2) for a small molecule; FGF-class risks (mitogenesis/tumour promotion, angiogenesis); regulatory plan (clinical-development dossier and clinical-trial application to the relevant national regulator(s); ethics review within the applicable human-research ethics system; an engineered FGF17 protein follows the biologicals pathway, whereas a gene-delivered FGF17 would be an advanced-therapy product and a GMO matter for the biosafety regulator).	Delivery solved; clean toxicology.
4 – human studies	Phase 1 safety/PK; biomarkers (MRI myelin water fraction/DTI, EEG, cognitive tests); structured learning input as digital training; Phase 2 in MCI/early AD and, separately, in neurodevelopmental populations (regulator and ethics-system scrutiny; assent plus proxy consent; independent data-safety monitoring).	Safety and biomarker signal.

Hard barriers that must be stated: no validated way exists to deliver FGF17 to the human brain; the mitogenic and angiogenic risk of an FGF-class CNS agent may never clear a paediatric risk–benefit assessment for a non-fatal indication such as a learning disability, and enhancement in people without disease carries a higher ethical bar than treatment of disease; 7,8-DHF is reported not to activate TrkB reliably in cell assays [2–4]; no human trials of FGF17 or 7,8-DHF were found (verify); healthy-animal enhancement may not transfer to disease or humans. The FGF23/phosphate effect is *not* the relevant class risk (a different, endocrine FGF).

What this study must deliver to earn Stage 1a: reagent activity and FGF17 pathway engagement (0a); a positive direction on the overall index (0b), with set-shifting reported separately; Stage 1a (the larger study) would additionally want an Olig2/MBP histology direction on the banked tissue, analysed with separate funding, and a structural effect is a gate for Stage 1b and 2. Covariation of structure with behaviour cannot be judged at this n.

3. Design and Animals

Young adult male Wistar rats (about 8–10 weeks old, ≈ 250 – 300 g at surgery) from an animal facility licensed under local animal-research regulations (a central animal facility or institutional breeder; strain and sanitary status stated; SPF Wistar supply, age-matched litters and price are confirmed directly with the facility). Males only. Housed on a reversed 12:12 light cycle so that dark-phase work falls in working hours, in perforated-divider cages in pairs of the same arm and time block (social contact with individual rations and less headcap chewing; single housing only with justification to the host institution's animal ethics/welfare committee, identically in all arms). The sponsor randomises by baseline weight and source litter and holds the allocation and drug-code key (section 10). The 30 main rats are split into two halves (A and B, 5 per arm each) that take the tasks in different orders, and into five fixed time blocks of six rats (two per arm, assigned at random within arm and kept for the whole study; each cage pair is one A rat and one B rat of the same arm and block). Blinding: an unblinded host technician (with a named backup) who does no testing or scoring prepares coded vials and syringes from the sponsor's key, wrapped in opaque sleeves with opaque-hub needles (7,8-DHF is a yellow flavone, so a clear vehicle or a clear syringe would reveal the arm); the injector and weigher never test or score; surgeon and behaviour operators are blind to drug; welfare and endpoint decisions are made by staff blind to arm. The environment arm cannot be fully blinded, so ELE handlers and test-day scorers are different people, set-shifting and Hebb-Williams are scored live by testers blind to environment where possible and verified from video under coded IDs, and 20% of videos are re-scored for inter-rater agreement (ICC). Arena identity is rotated by day within each arena type so that it is not aliased with arm.

Group	Content	Enrolled
Main cohort (one wave)	3 arms $\times$ 10: C (vehicle + standard), E (vehicle + ELE), D+E (FGF17 + 7,8-DHF + ELE); halves A/B of 5 per arm	30
Engagement satellite – FGF17	At 30 min after one ICV bolus (0.5 μ g in 3 μ l, as in the main cohort): time-matched vehicle 4, FGF17 4, heat-inactivated FGF17 (boiled aliquot, equal mass) 4; restraint duration logged; same tissue method in all groups; t0 = injector removal; punch of the periventricular wall (right hemisphere) and of the adjacent hippocampus (a descriptive peri-hippocampal readout) taken first, then the remaining slab and cannula track photographed (order recorded); pERK/total ERK; a satellite with confirmed misplacement is excluded from the gate ratio and replaced from the contingency rats	12
Practice/tolerability	2 cannula placement checks; 4 vehicle-tolerability rats (2 low-DMSO solution, 2 CMC-Na suspension, each receiving 7,8-DHF in its vehicle for the full 35 days; weekly weights, daily abdominal-wall palpation score, necropsy scoring). With 2 rats per vehicle this is a screen for gross intolerance only: >10% weight loss or any palpation/necropsy sign of peritonitis or adhesions in either rat triggers a host and sponsor review and a switch of vehicle. The chosen vehicle is a 35-day exposure in all arms, so IP sites and abdominal pain are also scored daily in the main cohort.	6
Pilot (unoperated)	4 unoperated rats used for the ED trial pace, exemplar salience, tester training and the IPTT-against-rectal swim calibration (transponders implanted under brief isoflurane; habituation, 3 weeks of restriction, training and testing in Phase 0a, weeks –8 to –4); included in the numbers submitted to the ethics committee	4
Total	Cannulated rats: $30 + 12 + 2 = 44$	52
Contingency (registered maximum, not enrolled unless needed)	One FGF17 re-test (4 vehicle + 4 FGF17 + 4 heat-inactivated, section 8) $\rightarrow$ registered maximum 64 rats, 56 cannulated	12

Enrolment is 10 per arm because ICV-cannula attrition in rats is commonly 30–40% (a planning assumption, not a sourced rate); the plan assumes 60–70% completion (6–7 per arm). Cannula loss is logged by arm. A **completer** is a rat that is alive with a patent cap through day 34 and has an observed score on the task in question; “observed score” is defined per task (a set-shifting score includes cap-scored rats that started testing, not non-starters). Because climbing and wheels exist only in the enrichment arms, cap loss is expected to be higher there: protected recessed caps with a smooth collar, climbing structures ≥ 15 cm below the lid and solid-surface (rung-free) wheels are used, an arm-stratified tipping-point analysis is pre-specified, the delivered FGF17 dose is reported per D+E completer, a completion difference of ≥ 3 completers between arms is flagged and reported (this will happen in about 45% of runs even with equal true completion, so it is descriptive only), and a difference of ≥ 5 triggers a review and a possible stop (about 8% false alarm under equal completion; because cap loss is expected to be higher in the ELE arms, some gap is expected rather than a pure false alarm). Singleton rats whose cage-mate was lost stay in their divider cage with the empty side open; days single-housed are logged by arm and used as a sensitivity covariate, and the ethics-committee justification for single housing covers this case. Welfare: the combined burden (cannula for ≈ 7 weeks, 35 daily injections, food restriction, swimming) is expected to be graded severe under the local severity classification, which typically includes moderate distress of long duration (the surgery alone would be moderate; the host confirms the labels; the ethics committee assigns the grade and the narrative argues for the lowest defensible grade with explicit refinements); a second rest day in the water maze or a repeated hypothermia is a veterinary-review trigger; the narrative gives a cumulative-severity justification and the refinements, with a scoresheet and humane endpoints: >15% weight loss from the growth-adjusted target, injection-site lesions, cannula infection, hypothermia

rule, neurological signs after ICV (any seizure: immediate euthanasia; circling or head tilt lasting >24 h: euthanasia), scoring daily and twice daily after surgery and during the restriction ramp, and a named growth curve (the supplying facility's Wistar growth chart) for the growth-adjusted target, weight loss of more than 20% of pre-surgery weight before restriction starts, a body-condition score during restriction, a palpation score threshold for IP peritonitis in the main cohort, a limit on repeated hypothermia, and the headcap (loosened with intact margins: one re-cementing under isoflurane; detached or exposed bone: euthanasia; decided by a veterinarian blind to arm, with the event logged by arm). Surgical, re-capping and terminal anaesthesia use isoflurane. The analgesia regimen is written by the host veterinarian (typically pre-emptive meloxicam daily for at least 72 h, dipyrone, and local lidocaine/bupivacaine at the incision, with antibiotic prophylaxis, eye lubricant, thermal support and fluids per SOP; post-operative scoring twice daily for 3 days, then daily; it is identical in all arms and fully washed out before the pERK gate and before behaviour); an opioid (buprenorphine or tramadol) is added if the ethics committee requires it or pain scores warrant it. Tramadol, buprenorphine, other opioids and ketamine are controlled substances under national rules and are used only through the host's authorised route (section 9). Surgery is limited to 10 rats a day on two frames with two surgeons (about 8 h each). Pre-specified exclusions: dye/histology-confirmed ventricular placement, swim speed <50% of cohort median (humane endpoints end a rat's participation but its earlier tasks are kept and the missing later tasks are handled by the tipping-point analysis); a rat excluded for a second water-maze rest day or a rest day on day 33 or 34 loses only the affected water-maze readout and keeps its other tasks. Failure to reach the set-shift training criterion is not an exclusion: it is a set-shifting non-starter (section 5). A satellite replaced after misplacement draws on the contingency rats first, and the re-test then draws on the remaining ones; if both occur, extra rats are requested from the ethics committee (maximum stated in the protocol).

Power (honest). With 6–7 completers per arm, a pairwise comparison reaches 80% power only for $d \approx 1.6$ – 1.8 (two-sample t-test). Healthy young rats are near ceiling on simple tasks, and a cross-species enrichment meta-analysis reports $g \approx 0.44$ in rodents (an unpublished dissertation [15]), so a null result is likely even if a modest true effect exists, and only a very large effect would be seen. Pilot effect sizes must not be used to power the next experiment from the point estimate [12,13].

4. Interventions

Component	Specification
FGF17 (days 0–6)	Recombinant mouse FGF17, one 50 µg pack (mouse–rat sequence alignment documented before ordering). Chronic guide cannula to the right lateral ventricle (scaled coordinates; a microinjection pump and Hamilton syringes are required). Four boluses on days 0, 2, 4, 6 in the D+E arm: 3 µl of ≈167 µg/ml (0.5 µg each; 2.0 µg per rat) in aCSF + 0.1% BSA (standard anti-adsorption practice, not a vendor specification; “carrier-free” FGF17 means no vendor-added BSA, which is distinct from this in-house carrier) + heparin 10 µg/ml, awake, 0.5–1 µl/min with the injector left 2–3 min; vehicle boluses in C and E. The 167 µg/ml working stock lies within vendor reconstitution ranges (typically ≥100 µg/ml, up to 1 mg/ml; reconstituted FGF17 keeps 2–7 days at 4 °C), so aliquots are frozen at –80 °C in carrier and used within about 4 months, recovery after reconstitution and freeze-thaw and a pERK potency recheck are done before the main cohort, low-bind tubes are used, the solution is inspected for precipitate, and if unstable the concentration is lowered and the volume raised (up to 5 µl; the shortfall in delivered mass is then declared); dead volume is measured and primed so that 3 µl is delivered. Supply from the single pack: D+E 10 × 2.0 = 20 µg; satellites 4 × 0.5 + heat-inactivated 4 × 0.5 = 4 µg; contingency re-test 4 × 0.5 + 4 × 0.5 = 4 µg; bioassay ≈5 µg (generous, including the boiled aliquot); freeze-thaw recovery loss and dead volume ≈10 µg; potency recheck ≈5 µg; ≈48 µg of 50 µg, so there is no margin and a 25 µg vial is the contingency. The pack is reconstituted in two lots; the working aliquots are used within about 4 months of reconstitution (the real exposure of the second lot is about 6 weeks), the potency recheck is repeated within 2 weeks of day 0, and a re-test (which adds 5–6 weeks) triggers the contingency vial. The boiled (heat-inactivated) aliquot is itself bioassayed (<10% residual pERK activity) and checked for aggregation. <u>Delivery gap and dose anchor:</u> Iram 2022 infused continuously for about a week with an Alzet 1007D (≈84 µl delivered); one reviewer retrieved the Methods as recombinant mouse FGF17 at 25 µg/ml (≈2.1 µg continuous) plus a separate acute 0.5 µg bolus in 2 µl; another reviewer could not retrieve it, so the host must confirm it in the Methods (PMC9377328 or the UCL copy) before ordering. Four 0.5 µg boluses (2.0 µg total, 12 µl) approximately match that one reported total mass but not the exposure time: they are pulsatile and ≈7× smaller in volume, so a null remains confounded with delivery and FGF17 stays exploratory.
7,8-DHF (days 0–34)	5 mg/kg IP daily, days 0–34 (35 injections; rotated sites) in the D+E arm; vehicle IP in C and E. Daily order: testing (if scheduled) → 7,8-DHF injection → ELE/control session 30 min later, so the agent is on board during enrichment (mouse half-life ≈2 h) and the next day’s testing falls at trough (≈18–24 h after dosing in mice; one oral rat report gives a terminal half-life of ≈23 h — if that is right the drug accumulates and testing is not drug-free). Rat precedent for 5 mg/kg IP daily includes 5 weeks in a rotenone model in 9-month-old rats [17] and an optic-nerve model with outcomes at 7–21 days [18]; mouse work exists [5]. The 35-day duration is at the long end of that precedent. Vehicle (decided by the tolerability check before the main cohort is ordered; the check is a screen only): default is a low-DMSO solution (5% DMSO / 40% PEG300 / 5% Tween-80 / 50% water per Selleck, or 10% / 40% / 5% / 45% saline per MedChemExpress/InvivoChem, ≈2.5 mg/ml; prepared fresh); fallback is an aqueous CMC-Na suspension (no peer-reviewed intraperitoneal PK found; an IP suspension may give erratic absorption, so a null could reflect exposure). Oil vehicles are avoided. Necropsy scoring of peritoneal adhesions/granulomas in all main-cohort rats, from coded photographs converted to greyscale or colour-normalised because 7,8-DHF is yellow and tints tissue (the residual risk is declared); identical vehicle in all arms.
Enriched learning environment (ELE)	Days 0–34, one 60-min solo session per day (the test rat alone; no partner rats, which removes social contact as a confound and the housing and handling of partner animals) 30 min after the 7,8-DHF injection, in the dark phase under red light [6,7]. This is a weak enrichment dose and a non-social one (declared). Minimum specification the host must meet regardless of local habit: arena ≥1 m ² (the same size for control), ≥3 novel objects per session, running wheels (solid surface) and tunnels/climbing in each ELE arena (arena identity rotated by day within type, so that it is not aliased with arm), ≥3 puzzle devices on a six-level ladder (latch, string-pull, lever, chew), small fixed-calorie rewards (counted per rat; deducted from the ration only once food restriction has begun), engagement minutes and ladder level logged per rat, an object library of ≥30 objects on a rotation schedule (“novel” means not re-presented to the same rat within 7 days), and a fourth ELE operator for cleaning and resets, objects are pre-staged in sets, and an object is not re-presented to the same rat within 7 days. On the first day of a rat’s set-shift tests the ELE/control session is replaced by a home-cage interval in all arms (dosing continues), so that the attentional set between test day 1 and day 2 is not differentially exposed to novel objects; that one lost session per rat is declared; between test day 1 and day 2 the drug can still act on consolidation of the attentional set in the D+E arm, so ED – ID2 is reported alongside ED as a descriptive measure, and only this interval is protected because it separates the two parts of the transfer test. <u>Excluded from the ELE: odour discrimination and digging puzzles.</u> Control arena: same duration and handling, empty. SOP with photos completed by the host before submission to the ethics committee.

5. Timeline and Workload (Day 0 = First Bolus, a Thursday; One Wave)

Days	Activity
-22 to -16	Arrival (a Wednesday) and 7-day acclimatisation; reversed light cycle.
-15 to -13	Cannula surgery (Wed–Fri) with analgesia, ≤10 rats/day on two frames with two surgeons; day -12 (Saturday) is a reserve surgery day (≥11 days recovery for a rat operated then); ≥12 days recovery otherwise.
-12 to -1	Recovery, handling, weighing; arena habituation from day -7 in an empty control-type arena (all rats; arm-neutral).
0–6	FGF17/vehicle boluses (days 0, 2, 4, 6; Thu, Sat, Mon, Wed).
0–34	ELE/control sessions and 7,8-DHF/vehicle every day (35 doses), weekends included; each rat has 34 ELE/control sessions because one is replaced on its first set-shift test day.
5–10	Gradual food restriction to 90–95% of the growth-adjusted target by day 10 (before task habituation begins on day 11); held through the evening of day 28 (humane limit <85%); each rat is rationed individually (divider cages) 2 h after the end of its own ELE/control session. Rewards in the ELE and bait in the set-shift and Hebb-Williams tasks are weighed and counted per rat and deducted under one rule. Hours since feeding at testing are recorded descriptively.
11–28	Tasks 1 and 2 run in parallel in two halves (A and B, 5 rats per arm each), in opposite orders. <u>Half A</u>: set-shifting days 11–22 (habituation 11–13; training days 14–18; tests start for 5 rats on each of days 19, 20 and 21 (start days are fixed by a Latin rotation, listed as C/E/D+E start days for blocks 1–5: B1 19/20/21; B2 20/21/19; B3 21/19/20; B4 19/21/20; B5 20/19/21, which gives five starters a day, one per block, with each arm’s starters over the three days as C 2/2/1, E 2/1/2 and D+E 1/2/2 (balanced overall, not per day); half B takes the half-A column of the next arm and adds 6 days: B1 26/27/25; B2 27/25/26; B3 25/26/27; B4 27/26/25; B5 25/27/26, i.e. five starters a day and C 2/1/2, E 1/2/2, D+E 2/2/1 over days 25–27) and last two consecutive days each: day 1 simple discrimination, compound discrimination and the first intradimensional shift (ID1); day 2 the second intradimensional shift (ID2) and the extradimensional shift (ED); last finish day 22), then Hebb-Williams days 23–28 (adaptation 23–24, then 4 problems on days 25–28). <u>Half B</u>: Hebb-Williams days 11–16 (adaptation 11–12, 4 problems on days 13–16), then set-shifting days 17–28 (habituation 17–19, training 20–24, tests start on days 25, 26, 27 last finish day 28). Set-shift rules: the training criterion is 6 consecutive correct on each of a simple odour and a simple medium discrimination (training exemplars distinct from test exemplars); rats that reach it hold on one maintenance session a day; a rat that has not reached it by the end of its training window (day 18 for A, day 24 for B) is a set-shifting non-starter (below); a day-before-test reminder is 4 trials with training exemplars; each arm gets 2, 2 and 1 starters over the three start days and the start day is recorded. Tests have a 2.25-h cap per rat per day and trials a 3-min limit and a 60-trial cap per stage. A stage unfinished at the day-1 cap resumes at the start of day 2 (trial counts accumulate toward the 60-trial cap, the consecutive-correct counter restarts, a no-dig trial at 3 min counts as a trial and an error in every stage, and no recap is given). Any carry-over stage and ID2 stop hard at 55 min of day 2; not reaching criterion by then, or within 60 trials, is a pre-ED stage failure; a rat that fails pre-ED is not given ED. A 5-min changeover follows, and ED then gets a budget of up to 75 min (ending at criterion) for every rat; 55 + 5 + 75 = 135 min fills the 2.25-h cap exactly, with no spare time, so the host confirms the timing in the pilot. Any rat that starts testing and does not reach the ED criterion for one of three behavioural reasons (pre-ED stage failure, reaching N_{cap} trials at ED, or time-out of the 75-min ED budget) is scored at the cap, log N_{cap}, in the index and in the fallback primary (a worst-score rule); a capped score is exempt from winsorising. N_{cap} is the number of ED trials that fit in 75 min at the pilot pace, at most 60; it is applied to the ED stage only (the pre-ED stages keep a 60-trial cap), and it is frozen in the OSF registration before surgery. A rat that fails the training criterion and so never starts testing is a set-shifting non-starter: it is reported as a separate completion outcome by arm, is missing from the set-shifting score in the primary, and is scored at the cap only in a sensitivity analysis. Death, humane endpoints and cap loss are never scored at the cap: they are missing data handled by the tipping-point analysis. In the set-shifting-alone rank/censored (Cox) analysis, time-outs and N_{cap} are right-censored at the trials reached and pre-ED failures are censored at N_{cap} + 1 (the worst rank); because ED pace may go with learning rate, a sensitivity analysis treats time-outs as failures at N_{cap}; rats that never started ED are excluded from the ED > ID2 validity check; the number of rats failing for each reason, stage times and time-outs are reported per rat and by arm, and an analysis that excludes them is a sensitivity analysis. The pace of ED trials is taken from the 4 unoperated pilot rats (or the host’s historical data); at about 1.25 min per trial a 75-min budget holds about 60 trials, but a realistic manual pace of 1.5–2 min holds only about 37–50, so the trial cap is set from the pilot to the number of trials that fit in 75 min, and frequent time-outs would trigger the time-out rule in section 6. There is no day 3, so Hebb-Williams or the water maze never overlaps an unfinished test. The overnight gap between day 1 and day 2 is a declared deviation from single-session protocols. Peak: 10 rats in set-shift testing in one day, but at most 4 testing at the same moment (two overlapping blocks of at most 2 each). Each block also holds 3 half-B trainees (one per arm, about 1 h each) and, on days 19–20, half-A rats needing a maintenance or reminder session, so each block window needs about 2 trainers running sequentially and about 4 trainers during overlaps: roughly 4 testers plus 4 trainers, i.e. about 8 people at once and about 10 rostered to cover statutory breaks, and at most 10 digging boxes at once (4 test, up to 4 training, 1–2 maintenance, sequenced). Room 1 is for tests only (at most 4 boxes); room 2 is for training, maintenance and reminders (at most 5 at a time), so that training odours do not enter the test room. On days 25–28 there are 4 testers and no trainees, half-B maintenance and reminders on days 25–26, and about 2 separate operators for half-A Hebb-Williams; at most 5 digging boxes per ventilated room (two rooms) to limit odour cross-contamination. With lights-off at 07:00 the clock times are: behavioural starts 07:00, 08:15, 09:30, 10:45 and 12:00; injections 09:30, 10:45, 12:00, 13:15 and 14:30; ELE sessions 10:00, 11:15, 12:30, 13:45 and 15:00; rations 13:00, 14:15, 15:30, 16:45 and 18:00. The host’s per-day, per-block timetable for days 19–22 and 25–28 is part of the SOP pack. Testers rotate across arms; tester identity, start day and days of maintenance are recorded and used in a sensitivity analysis; the same tester runs a rat on days 1 and 2. In half B the odour/medium dimension order is flipped so that each arm has each first dimension 5:5 over both halves (3:2 within an arm and half), assigned independently of start day and recorded. The cage-mate of a home-cage rat returns from the ELE with object odours through the divider; this minor exposure is declared. Food restriction ends on the evening of day 28.
Days	Activity
29–34	Task 3 (last; non-appetitive): water maze for all 30 rats, run one block at a time (a block’s six rats swim interleaved, 4 trials each, in about 40 min, inside their pre-dose window) — acquisition days 29–32, probe day 33, visible-platform test day 34 (Appendix A). It

Days	Activity
	runs through one weekend (two operators, as in Appendix B). Ad libitum food resumes on the evening of day 28; the dummy caps are sealed and inspected after each session, swim speed is reported by arm, a rat that needs one rest day (a temperature below the device-calibrated threshold) has the rest of its acquisition shifted by one day (last acquisition day 33) and then runs probe and visible-platform test on day 34 at least 1 h apart; a second rest day, or a rest day on day 33 or 34, means exclusion and a flag; temperature is taken with subcutaneous transponders (IPTT) implanted under the surgery anaesthesia (calibrated against paired rectal readings across swims in the pilot rats before the study: an IPTT reading below the calibrated threshold at the end of a session triggers a rectal reading within 2 min, and a rectal reading below 34.5 °C means a rest day; the rat is then rewarmed and re-checked), not repeated rectal measurement; distal cues are large and high-contrast for albino rats at 10–20 lux.
35–36	Under anaesthesia an ICV dye injection (3 µl) is made where the cannula is patent (a blocked cannula is recorded; placement is judged from the cannula track, and resistance and backflow are recorded at every bolus), then decapitation (whole cage pairs, 8 + 7 pairs a day as balanced by arm as possible, with two teams, an extra tissue person per team, ≤15 min per animal including the peritoneal necropsy photographs, balanced by arm and block; day 37 is a reserve day); a 2-mm coronal slab through the cannula track is cut fresh from the right hemisphere and photographed; the rest of the right hemisphere is snap-frozen and banked at –80 °C; the left hemisphere is cut into 1.5–2 mm slabs, fixed 48 h, cryoprotected in sucrose and banked as embedded blocks at –80 °C (or in antifreeze at –20 °C). Banked tissue is not analysed in this study.

Daily schedule and workload. The 30 rats are in five fixed blocks of six (two per arm). A rat's sequence is: behaviour in its pre-dose window (set-shift test ≤2.25 h; training ≈1 h; Hebb-Williams ≈0.5 h; water maze ≈0.7 h for the block) → 7,8-DHF/vehicle injection → ELE/control session 30 min later (60 min) → individual ration 2 h after the session ends (on a rat's first set-shift test day, in which the session is replaced by a home-cage interval, at the same clock time as on the day before). Block injections are at 2.5, 3.75, 5.0, 6.25 and 7.5 h after lights-off, and behavioural starts are staggered by 75 min per block, so each block has exactly 2.5 h from its behavioural start to its injection (a 2.25-h set-shift test, a hard cap that never shifts a block's start, leaves 15 min for transport and set-up); the ICV bolus on bolus days is given within the pre-dose window, before the IP injection, and at the same point for every arm; each rat's ELE session starts 30 min after its own injection (+0/–5 min tolerance), logged, in a fixed order of rats into arenas so that every arena gets a 15-min turnaround; a second physical set of objects is pre-staged so that objects are not washed between blocks, and if the turnaround proves too short the host adds two spare arenas (≈USD 0.3–0.6k) or shortens sessions to 55 min; the last ELE session ends 9 h and the last ration falls 11 h after lights-off, inside the 12-h dark phase. Block b starts at lights-off + 75 (b–1) min, so about 4 testers and 4 trainers are needed at once (about 8, rostered as about 10). Hebb-Williams and the water maze use dim white light (≈10–20 lux, rooms only), a declared exception to the red-light rule (albino rats see little under red light). Six arenas are used (four ELE, two control), one operator per two arenas plus a fourth ELE helper for resets, so the working day spans about 13 h (lights-off minus 1 h to +12 h) covered by two staggered shifts; the host states the clock times of lights-off and the roster (days –15 to 37, about 53 consecutive days including weekends), with a backup for the unblinded technician. Ordinary days need about 30 operator-hours (about 6 people, including four ELE staff); set-shift test days about 67, or about 75 with maintenance and the technician (about 12–15 people at the busiest hours, with ELE staff, the blinded injector and weigher, and testers kept separate; ration and weighing staff do not score). Day 0 is a Thursday so the peaks fall on Tuesday–Friday and Monday–Thursday; the weekend rota repeats the same schedule. In-life: arrival day –22 to euthanasia day 36 = 59 days (8.4 weeks). Phase 0a (reagents, tolerability, satellites) comes first; its calendar is in section 9. The fixed task orders (A: set-shifting → Hebb-Williams; B: Hebb-Williams → set-shifting; water maze last for both) are declared and the half is a fixed effect in the analysis; ELE and DHF after testing form a post-training consolidation window, declared. There is no fallback schedule: if the host cannot name the roster, the study does not start.

6. Endpoints and Tissue

Primary analysis. The three task scores are standardised with the grand mean and the pooled within-arm SD estimated in-sample (no prior data exist), winsorised on residuals at ± 3 , signs aligned. They are analysed in one linear mixed model (task as a repeated factor; random intercept per animal; time block as a random intercept if it does not collapse to the boundary; half (task order) as a fixed effect; arm $\times$ task terms; baseline weight as the only covariate; common residual variance, with task-specific variances if the fit allows), using all animals with ≥ 1 task, with Kenward-Roger intervals (SAS PROC MIXED with DDFM=KR is the primary software; R packages are used only as a sensitivity check, because KR with heterogeneous variances is not available there). The gate statistic is the index contrast (mean of the three z-scores) $D+E - C$ in per-task-SD units, with $D+E - E$ and $E - C$ reported; set-shifting alone and the index without the water maze are reported separately. Sensitivity analyses: complete cases; tipping-point (δ -adjusted) analysis for informative dropout (the mixed model is valid under missing-at-random only). Wheel running is zero in arm C and post-restriction weight is post-randomisation, so neither is a covariate. Because ~ 18 – 21 completers across three tasks may not support the full model, a simpler fallback primary is pre-registered: per-task contrasts ($D+E - C$, $D+E - E$, $E - C$) with t-based 95% intervals, combined as the unweighted mean of the three standardised contrasts. The index and this fallback always use the capped log set-shift score (every rat that starts testing and does not complete ED for a behavioural reason, at the cap, section 5); the rank/censored (Cox) analysis is a separate set-shifting-alone readout. Because ED pace may differ by arm, differential time-outs can move the index (a reviewer simulation found that a 30-point difference in time-outs between arms moves the set-shifting contrast by about 0.5–0.6 SD and the index by about 0.2 SD, and raises the null false-go from 10% to about 20%; with 20% non-completion in every arm the target go rate falls from 72% to about 65%); the number of time-outs by arm is reported. A single rule governs this: the set-shifting-alone primary is always the rank/censored (Cox) analysis, and the index is always reported with and without set-shifting; the direction gate is applied to the index, and a “go” is only provisional unless the ED-completers-only set-shifting contrast has the same sign ($D+E \geq C$ and $D+E \geq E$), in which case it stands; if the signs differ, the outcome is “redesign”, not “go”. Time-outs that differ between arms by 2 or more rats (about 30–40 percentage points with 5–7 starters per arm) are flagged. Because chance alone would trip a data-driven trigger in about a quarter to two-thirds of studies at a realistic 10–20% time-out rate, no analysis is switched on a trigger. The operating characteristics below assume no non-completion and are void if time-outs are frequent. The inter-task correlation matrix is computed within arm from residuals and is descriptive. Analysis is run on coded labels by an analyst who does not hold the key (a third-party statistician if possible) until data lock.

Set-shifting score. Criterion 6 consecutive correct, maximum 60 trials per stage; the primary score is $\log(\text{trials to criterion at the extradimensional shift, capped at } N_{\text{cap}} \leq 60)$, with rats that start testing and do not reach it for a behavioural reason assigned the cap (section 5). The index and the OC table below assume the capped log score and no non-completion; the validity check ($ED > ID2$) is run only on rats that started ED, with time-outs and the 60-trial cap right-censored at the trials reached. The worst-score rule also makes selection from differential cap loss larger (a reviewer simulation: null false-go about 16% instead of 12%), and the tipping-point analysis does not cover worst-scored rats, so an analysis that excludes pre-ED failures (and, separately, one that caps non-starters), stratified by arm and by when cap loss occurred, is also reported. The set-shifting score mixes attention with pace, motivation and appetite, so an ED-completers-only transfer check is reported next to the index. Conditioning on the post-randomisation ID2 score is a sensitivity analysis only; $ED - ID2$ is reported as a descriptive measure. Water maze: mean proximity to the platform in the 24-h probe [21] (start-position correction stated by the host); quadrant time also reported.

Block	Readouts	n
Behaviour	Three tasks; scored live and verified from coded video.	all completers
Banked tissue (every main-cohort animal)	Left-hemisphere slabs fixed and banked (embedded blocks at -80°C or antifreeze at -20°C); right hemisphere frozen. No analysis in this study; reserved for Olig2/MBP histology and synaptic markers in the larger study (no EdU label is given in this study; immersion fixation gives lower and depth-dependent quality than perfusion — declared).	30
Engagement (satellites, snap-frozen)	pERK/total ERK in periventricular tissue (gate anchor) and the adjacent hippocampus (descriptive, peri-hippocampal); ice-cold dissection (<2 min), phosphatase inhibitors, FGF2-stimulated positive-control lysate and pre-specified assay CV on every blot; the same killing and dissection method in every group, which makes any stress- or anaesthesia-related change in pERK common to all groups (decapitation without anaesthesia needs justification to the host institution’s animal ethics/welfare committee under local euthanasia guidelines; published comparisons find isoflurane and unanaesthetised decapitation similar, and both can raise stress-related ERK signalling); one gel carries all 12 samples plus the bridging control so fold-changes are compared on one membrane; pERK and total ERK on the same membrane (strip and re-probe, validated) or a parallel gel.	4 + 4 + 4

7. Statistics and Pre-registration

Register on OSF before surgery (sponsor does this); ARRIVE 2.0 [14]; flow diagram of enrolled, excluded and completed animals. Estimation only: model-based contrasts (adjusted only for half and baseline weight; the fallback per-task t-intervals are unadjusted) with Kenward-Roger (KR) 95% intervals as primary; an animal-level (cluster) bootstrap that re-runs standardisation, winsorising and the model in every replicate is reported as sensitivity; no significance claims are made. The task order halves are balanced by arm, so the main contrast is protected, but arm $\times$ half is not usefully estimable. Sizing of the next study: use the upper 80% confidence limit of the pilot's raw-unit residual SD per task [20] and the full factorial/interaction design.

8. Pre-registered Decision Rules and Operating Characteristics

Phase	Pass condition		
0a – reagents	FGF17 bioassay active on the pack; mouse–rat alignment documented; DHF vehicle stable and tolerability screen passed (switch rule applied); potency recheck passed before the main cohort.		
0a – FGF17 engagement (gate; signalling at the ventricular interface — necessary, not sufficient; the hippocampal punch is a descriptive peri-hippocampal readout, since the dorsal hippocampus abuts the ventricle)	Periventricular pERK/total ERK, ratio of group means, FGF17 $\geq 1.5 \times$ time-matched vehicle AND FGF17 $\geq 1.25 \times$ heat-inactivated FGF17, at 30 min (n = 4 vehicle, 4 FGF17, 4 heat-inactivated; point estimates; acceptance CV pre-specified; positive-control lysate on every blot; the threshold applies to the diluted punch tissue actually assayed). A borderline result (FGF17/vehicle ratio 1.3 to <1.5) triggers one re-test (fresh 4 + 4 + 4 rats at 30 min, same pass rule; the decision rests on the fresh sample alone). Operating characteristics (200,000 simulations; lognormal assay noise at CV 30% / 50%; effect present; heat-inactivated group at vehicle level): false-pass with the re-test $\approx 1.6\%$ / 7.8% if there is no effect; pass $\approx 26\%$ / 30% at a true 1.3-fold change; $\approx 57\%$ / 49% at 1.5-fold; $\approx 97\%$ / 83% at 2-fold. A true 1.5-fold engagement therefore passes only about half the time: the gate detects large engagement and can miss a real but modest one. These figures assume the heat-inactivated group sits at vehicle level and assay CV $\leq 50\%$: if boiling leaves partial activity (heat-inactivated at 1.5 $\times$ when the true effect is 2 $\times$) the pass rate at 2-fold falls to $\approx 64\%$ / 56%, and at assay CV 70% the pass rate at 1.5-fold falls to $\approx 45\%$ with a false-pass of $\approx 13\%$ (round-1 re-simulation by a reviewer). The heat-inactivated ratio is therefore the limiting criterion, and the boiled aliquot is bioassayed. Satellites are fed ad libitum and recently cannulated, so engagement shown there may not equal engagement in the later restricted cohort, and the stress of awake restraint raises baseline pERK in all groups, which compresses the fold-change and makes the gate harder, not easier, to pass. The single 30-min time point is an assumption anchored only by the FGF2 control curve and an FGF1 ICV precedent in hypothalamus (pERK detectable from ≈ 20 min and sustained for hours; a different ligand and region, so an extrapolation to FGF17 at the periventricular wall); if the true time course is different the gate can miss it. FGF17, an FGF8-subfamily ligand, is expected to signal through c-isoform FGFRs (FGFR1c/2c/3c and FGFR4, with a preference for FGFR3c/FGFR4, per Zhang 2006 [23]; FGF17-specific profile to be verified); Fortin 2005 [16] tested FGF2, FGF8, FGF9, FGF17 and FGF18 on oligodendrocyte-lineage cells; FGF8 and FGF17 acted through FGFR3 and, as reported, inhibited the final differentiation of precursor cells, a subset of the responses to FGF2, which is the opposite direction to the pro-precursor effects reported by Iram 2022, so the net effect of FGF17 on these cells is not settled (secondhand reports; to be checked against the paper). If the gate is not met, the main cohort does not start; test a higher dose or continuous infusion in a small follow-up. An FGFR-inhibitor arm is deferred to the follow-up experiment.		
0b – feasibility	$\geq 50\%$ of the 30 enrolled rats (≥ 15) have an observed score on all three tasks (a rat scored at the cap after starting testing, including a pre-ED failure, counts as observed; a set-shifting non-starter does not, and cap-scored rats are reported separately); cannula placement $\geq 75\%$ correct in the main cohort (a different test from the host-competence check of ≥ 12 of 14 in section 10, the smallest whole number that reaches 80%). (Binomial P($\geq 15/30$): 0.77 at 55%, 0.90 at 60%, 0.97 at 65%, 0.99 at 70% true completion; a single wave, so no wave clustering.)		
0b – direction	Index contrast D+E – C $\geq +0.5$ per-task SD (in-sample pooled SD; a small circularity that is declared) and D+E $\geq$ E in point estimate. The gate applies to the point estimate; the interval is reported alongside and is not itself a gate. A reported, non-gating transfer flag records whether the set-shifting contrast D+E – C and D+E – E are both ≥ 0 (adding it as a requirement would lower the target-scenario direction-only go rate from about 72% to about 62%).		
Provisional go	Phase 0a passed AND 0b feasibility AND 0b direction. Joint probability, illustratively assuming independence and 90% feasibility: $\approx 63\%$ (the joint probability of gate, feasibility and direction, not the 72% direction-only figure (62% with the transfer flag) above; with 20% non-completion in every arm it falls to about 57%) if FGF17 truly engages at 2-fold and the target effect holds, $\approx 37\%$ if engagement is 1.5-fold; $\approx 10\%$ if FGF17 engages but behaviour is null; $\approx 0.2\%$ under a true global null (CV 30%; larger at CV 50%). Differential cap loss can select fitter D+E completers and bias the direction upward; the arm-stratified tipping-point analysis addresses this.		
Operating characteristics of the direction rule (simulated; 6 completers per arm 7 completers per arm; three tasks; index contrast; r = inter-task correlation; percentage that would be a “go” on direction alone):			
True situation	r=0	r=0.3	r=0.6
No effect	7% 5%	11% 9%	13% 12%
Modest (E 0.3, D+E 0.5)	45% 45%	43% 43%	41% 42%
Target (E 0.3, D+E 0.8)	79% 81%	72% 74%	67% 69%
Large (E 0.5, D+E 1.2)	97% 98%	92% 94%	87% 89%
Enrichment alone (E 0.6, D+E 0.6)	39% 40%	39% 38%	38% 38%

The rule is lenient (about 10% false-go) and cannot tell the drug package from enrichment: enrichment alone gives a “go” in about 39% of runs. The table assumes the same effect on all three tasks: if the effect is on set-shifting only (the clean transfer readout) the go rate is only about 25% at the target size, and if it is on the two spatial tasks only about 48%, so the index can be driven by the spatial tasks and miss a transfer-specific effect ($r = 0.3$, 6 completers). It only screens for a promising direction. A “go” justifies the larger study only; a “no-go” identifies the failure mode (delivery, engagement, ceiling, attrition) and is not evidence that the approach cannot work.

9. Logistics and Budget

Item	What the review found
Ethics	Animal use is approved by the host institution's animal ethics/welfare committee under local animal-research regulations. The host institution and its animal facilities must hold any registration or accreditation that local animal-research regulations require; check that it is current before choosing a host, because a lapsed accreditation bars animal work. Invasiveness is graded under the local severity classification (the cannulation surgery alone is likely to be moderate, but the cumulative burden of about 7 weeks with an implanted headcap, 35 daily injections, food restriction and swimming is likely to be classed severe, which some classifications define to include moderate distress of long duration; plan for the longer review that a severe grade brings, and the host confirms the current labels). Every person on the protocol needs documented training as local regulations require (about 30 h is commonly required, covering ethics, welfare and procedure-specific training). Review time differs by institution (1–6 months; a reviewer assumption); ask for an interim review. The principal investigator must be host staff. Animals are reserved only after approval; practice and contingency rats are in the approved protocol; the researcher reports to the committee on the host institution's form, the committee makes any reports the local regulator requires, and carcass and biological-waste disposal follow local rules.
Imports	FGF17, the antibodies, the cannula sets, the 7,8-DHF and the transponders come from different vendors, so plan four import shipments (or a local distributor, whose mark-up can be large): about 4–12 weeks each including licensing, customs and cold chain (reviewer assumption), customs-broker fees, dry-ice courier. Import licences, any tax relief or annual quota for research imports, and clearance of controlled or animal-origin items depend on national import, customs and controlled-substance rules; the host confirms them with its import office or the local regulator. Import duties and fees vary by country and are not included. Base case: all imports (Phase 0a and 0b items) are licensed and bought in one early window, normally before ethics approval , because licences or quota may not be available later; this accepts about USD 1.0–1.5k of sunk cost if the gate fails (cannulas, 7,8-DHF top-up and transponders). If the committee refuses or delays, the imported lines (about USD 3.5–5k landed, including the IPTT transponders) are at risk; if the host will not buy before approval, imports go to distributor pricing or wait. Keep a fallback: a direct import by the host institution or a local distributor. If a local distributor at 2–3× list must be used, the imported lines cost about USD 6–9.5k in total (unverified). BSA, FBS, animal-derived secondary antibodies and the 3T3 cells for the bioassay are sourced locally where possible (for example from a local cell bank), because animal-origin imports may need separate prior clearance; that clearance step is added to the calendar if an import is unavoidable. FGF17 and antibodies are carrier-free where possible. Each shipment may need its own licence, so a split or back-ordered shipment such as the contingency vial may need a new licence and broker fee (about USD 0.25–0.4k each); confirm whether the licence must precede payment. Purchases follow the host's procurement rules (quotes, a sole-source justification for single-vendor FGF17, a handling fee per import), which adds weeks to the order calendar. The base budget assumes landed cost ≈1.3–1.5× list; the four-shipment line (USD 1.0–1.5k) covers broker, courier and cold-chain fees only, and handling, airport storage and per-process fees may add USD 0.3–0.6k (within the pessimistic overrun). The antibodies for the pERK/total-ERK bioassay must arrive by week –14 of the Phase 0a calendar, not only by the westerns at week –8. Local regulatory, import and price details were not verified; the host must confirm them.
Controlled drugs	Ketamine, buprenorphine and other opioids, tramadol and, in some places, xylazine are controlled under national controlled-substance rules. The host states its route (for example veterinary products bought and prescribed by a licensed veterinarian, or human-use products under a special authorisation from the local regulator); some authorisations cover strictly academic use only, do not authorise import (a separate import authorisation may be needed) and require movement reports. Every route needs a locked store and a stock register; the host checks the current controlled-substance schedules. Many ethics committees expect an opioid for stereotaxic surgery, so this route is likely to be used. This protocol plans isoflurane for anaesthesia and meloxicam, dipyrone and local infiltration for analgesia, and uses controlled drugs only through the host's authorised route.
Equipment access (precondition)	The budget assumes written access to: stereotaxic frames (two), a microinjection pump, a water maze with tracking software, a cryostat or vibratome only if banked tissue is later processed, a western blot system and image-analysis software. Multi-user facilities often charge hourly or require a cost contribution; those fees are NOT in any scenario here and must be quoted. The ELE arenas, Hebb-Williams mazes and digging boxes are priced as fabricated in the base case; the optimistic case assumes the host owns them.
Personnel, services, overhead	Not in the budget. About 2,550 operator-hours (Appendix B), a part-time coordinator over about 9–14 months, a surgeon, a veterinary lead and an unblinded technician. Operator cost depends on how people are engaged (for example students on stipends, salaried technicians or contract staff) and on local employment and stipend rules; weekend and night premiums are part of the unquoted personnel cost. Unverified estimate: operators about USD 6–26k for about 2,550 operator-hours (stipend-level at the low end, salaried technicians at the high end). Adding a part-time coordinator (USD 3–10k, covering about 9–14 months of part-time work, stipend-level at the low end and a salaried part-time coordinator at the high end), other staff and equipment fees (USD 1.5–4k), a materials cost of about USD 9–15k and institutional overhead (host-specific; reviewer examples 5–20%), the plausible all-in total is about USD 25–65k (an unverified estimate; the host must quote; the low end assumes stipend-level operators and about 5% overhead, rounded up for the unquoted items, and the high end salaried staff and about 20% overhead; adverse scenarios and premiums could add more). Student stipend rules, dedication limits, night and weekend access to the animal facility and thesis/authorship expectations must be agreed with the host.
Timeline	Contracting and the work plan (2–6 months, not counted); ethics review 1–6 months; with FGF17 ordered at ethics approval, Phase 0a runs about 24 weeks before the cohort arrives (calendar below), then 8.4 weeks in-life and about 1 month of analysis: about 9–14 months from ethics submission to core results (about 7 months if FGF17 is ordered and bioassayed before approval, which needs the host to accept the risk). A triggered re-test adds 5–6 weeks. Institutional risks (strikes, closures) are a host-selection criterion.

Phase 0a calendar (planning offsets in weeks before the main cohort arrives on day -22; the host confirms): Ethics approval and FGF17 order (-24); pack arrives, bioassay of the pack and the boiled aliquot (-14 to -12); tolerability rats start dosing (-14) and finish their dosing day 35 (-9); satellite and practice surgeries (-12 to -11) with dosing and tissue at -10 to -9; westerns and gate readout (-8 to -6); written decision (-5, with the placement score of all 14 satellite and practice rats); main-cohort rats ordered (-4, after the gate and the vehicle decision; age-matched males need provisional reservation at ethics approval); the second FGF17 lot is reconstituted about 5 weeks before day 0 so that no aliquot is older than about 4 months at the last bolus. Satellite and pilot surgery never shares a day with main-cohort surgery.

Budget (USD ranges; list prices from search plus reviewer knowledge; to be replaced by quotes):

Line (core = Phase 0a + 0b)	Base	Notes
Rats: 52 (30 main, 12 satellites, 6 practice/tolerability, 4 unoperated pilot rats) at external-rate prices, with transport	0.9–1.3k	Facility price lists vary by site (not verified; ask the supplying facility to quote). Internal rates apply only to internal users.
Housing/care (≈2,250 rat-days ≈ 1,100 divider-cage pair cage-days, reversed-cycle room)	0.8–1.2k	Local per-diem not found (ask the facility); one published price list gave USD 0.47–0.94 per cage-day (used only as an order of magnitude); range about USD 0.55–1.4k; doubles if pair-housing is refused
FGF17, one 50 µg pack, landed	0.5–0.8k	Cohesion ≈ USD 480/50 µg via an aggregator; R&D; 25 µg mouse vial price not found; PeproTech lists human only; a 25 µg contingency vial (USD 0.35–0.55k) in the pessimistic case
7,8-DHF (purchase ≈2 g for pack size and margin; need ≈1 g) and vehicle components, landed	0.2–0.3k	TCI ≈ USD 62/g
Cannulas, guides, dummies, injectors, screws, cement (44 rats), microinjection tubing	0.65–1.0k	The base assumes a mix: commercial sets ≈ USD 20–30 each landed (USD 0.9–1.3k for all 44), home-made guides from steel tubing ≈ USD 0.3–0.5k; divider cages, split hoppers, an IPTT reader and about 35 transponders are in small consumables
Anaesthesia, analgesia, surgical and terminal consumables (44 cannulations, 48 terminal procedures), including controlled-drug handling	0.4–0.6k	
ELE: 6 arenas (4 ELE, 2 control) ≥1 m ² , solid wheels, tunnels, objects, six-level puzzle ladder	1.5–2.2k	Fabricated locally ≈ USD 150–300 per arena; get two quotes; USD 0.3–0.5k if the host owns the arenas
Behaviour: 2 Hebb-Williams mazes, ≥10 digging boxes in two rooms, set-shift pots, media and odours (≥10 exemplars per dimension)	0.8–1.2k	0.4–0.6k if the host owns the mazes and boxes
ERK block, landed: pERK, total ERK, HRP secondaries, loading control, FGF2 positive control	1.3–1.9k	list ≈ 1.0–1.4k × 1.35
Western reagents (membranes, ECL, inhibitors)	0.3–0.45k	
Four import shipments (FGF17; ERK antibodies and western reagents; cannulas and 7,8-DHF; IPTT transponders), each with its own import licence or permit, customs-broker, courier and cold-chain fees	1.0–1.5k	≈0.25–0.4k per shipment; consolidating through one forwarder or distributor, or buying 7,8-DHF locally, would cost less; local broker and storage fees not quoted; import duties and fees vary by country and are not included
Small consumables: Hamilton syringes, aCSF, BSA, heparin, low-bind tubes, opaque sleeves, 3T3 cells and culture, food/reward pellets and bedding if billed, cameras (infrared) and storage, red/white lights, perforated cage dividers, tempera paint, dye, fixative, sucrose, cryovials, waste, dry ice	1.0–1.5k	Reviewer estimates range 0.8–1.5k; an IPTT reader (≈USD 1–2.5k) and 4–6 infrared cameras are assumed to be host-owned equipment, and cost about 1.5–3.5k if they must be bought
IPTT transponders (≈35, landed ≈USD 15–18 each) and opaque blinding sleeves and needles	0.5–0.75k	Imported; quote only; the reader and infrared cameras are assumed host-owned (1.5–3.5k if they must be bought)
Ethics-committee fee and miscellaneous	0.05–0.15k	
Contingency rats (12 registered re-test rats: purchase, housing, surgery)	0.4–0.6k	Spent only if the re-test is triggered
Core base total, materials only	9–15k	excludes personnel, overhead, equipment fees, FX reserve
Scenario	Core	Assumptions (itemised)
Optimistic	≈ 8–11k	host owns arenas, mazes and boxes (about -1.5 to -2.3k); host animal facility at internal rates (rats about -0.3 to -0.4k, housing about -0.35 to -0.55k); a host that already holds the total-ERK and loading-control antibodies (about -0.1 to -0.2k); home-made cannula guides (about -0.3 to -0.5k); shipments not consolidated
Base	≈ 9–15k	as above

Scenario	Core	Assumptions (itemised)
Pessimistic	≈ 12.5–18k	15% overrun plus a contingency 25 µg FGF17 vial; equipment fees, personnel and overhead still excluded
Pessimistic plus import duties and taxes (no import exemption)	≈ 14–20k	imported lines taxed (+1.2–2.2k on about 3.0–3.4k customs value, unverified; import duties and fees vary by country); also likely months of delay
Exemption unavailable and a local distributor used (2–3× list)	≈ 15–23k	imported lines ≈6.2–9.3k instead of ≈3.5–5k (+2.0–5.1k) on top of the pessimistic case (conservative: the overrun is applied twice and the broker line is kept)

Materials budget: the planned materials fit a budget of up to about USD 15k. The optimistic case (≈8–11k) stays inside it even with a 10–15% currency reserve; the base (≈9–15k) fits it before the reserve (≈0.9–2.3k, not in any line), which would push the base to about 10–17.5k, so the upper end of the base can exceed it; the pessimistic cases (≈12.5–18k, ≈14–20k and ≈15–23k) can exceed it. This protocol counts the reserve against the budget; a budget of about USD 15k covers the base with the reserve in most cases but not every triggered re-test, pessimistic or import-duty scenario; personnel and overhead remain outside the materials budget, and import duties and fees vary by country and are not included. The reserve is applied to the whole base as a deliberately conservative figure; on locally priced spend alone (about 7–10k) it would be about 0.7–1.5k. A triggered re-test costs a further ≈0.8–1.3k (a 25 µg vial ≈0.35–0.55k, price not found, two 10 µg vials at about USD 270–380 are an alternative; plus ≈0.3–0.45k for the extra import and ≈0.15–0.25k for a second western run), and it adds about 10–18 weeks (a further import of 4–12 weeks plus 5–6 weeks of work), or only 5–6 weeks if the vial is bought in the Q1 window (sunk if unused); the base includes the 0.4–0.6k contingency-rat line, which is a floor, but not the vial, so a triggered re-test in an otherwise-base year totals ≈10–16k (≈11–18k with the reserve); the pessimistic scenario already includes the vial. Phases: 0a ≈ 5–7.5k (rats, housing, FGF17, DHF, cannulas, surgical, ERK, western, imports, small consumables, ethics-committee fee and contingency rats), with a stop rule at that spend; 0b ≈ 5.4–7.7k (rats, housing, DHF, cannulas, surgical, ELE, behaviour, small consumables and IPTT transponders). By use, housing splits about one quarter (0a) / three quarters (0b) (the phase figures above front-load it for cash timing); the 7,8-DHF, the main-cohort cannula sets and the transponders are imported in the same first-quarter window, so the cash committed before the gate is nearer 6–8.5k (the 0a figure less the unspent contingency rats, plus about 1.0–1.5k of 0b imports) plus 0a personnel. First cuts if prices run high: use host-owned apparatus; home-made cannula guides; do not register the contingency rats (a smaller arm size would need the block design redone and is not recommended). The western block must not be cut (it carries the only gate). Must not be cut: placement check, FGF17 bioassay and alignment, blinding/randomisation, DHF tolerability screen, visible-platform control, banking of tissue.

10. Lab Handover and Remote Oversight

The researcher will not necessarily be on site; the host site runs the experiment. This PDF is the working protocol; the following must exist before animals arrive. SOPs, ethics-committee forms and scoresheets are kept bilingual where needed (the staff's working language; English for the sponsor).

Item	Responsibility / requirement
Legal form (decide first)	Either (a) an academic or research partnership with the host institution, usually with funds received and managed by the host or its designated research-support body as intervening party; the overhead is set by the host; IP and data ownership are written into the instrument (shared per the agreement) and the host's technology-transfer office is usually involved; or (b) a service contract with an invoice and taxes agreed; results then usually belong to the contracting party, which the contract must state; any payment to students under this route must be confirmed with the host. Route (b) probably loses several assumptions of the base budget at once (research-only import exemptions, controlled-drug authorisations limited to academic use, and student stipends), so route (a) is the route consistent with the base budget; route (b) must be re-costed with full taxes and staff pay. Which route pays which people is written into the instrument. The sponsor is an individual: the protocol must name the legal entity through which the money enters (a sponsor company, a foreign institution, or a donation), and the host's legal office must confirm it can contract with that entity before this step. The sponsor's residence decides currency exchange, transfer taxes and fees, withholding and, for a donation, any donation tax; these vary by country and must be stated; whether goods imported under a research exemption may be used in a paid service contract must be confirmed with the local regulator. Foreign funds arrive by international transfer with the usual exchange documentation; confirm in writing with the host that foreign funds can be received for this kind of contract. The choice determines payment mechanics and is a precondition for submission to the host institution's animal ethics/welfare committee.

Item	Responsibility / requirement
Host selection and due diligence	Criteria: a host site (university, clinic, hospital research unit, contract research organisation or other facility) with a licensed animal facility, a functioning animal ethics/welfare committee and approval under local animal-research regulations; a named route for controlled drugs under national controlled-substance rules, with a locked store, a stock register and a registered veterinary lead; an importer of record able to import research reagents under national import, customs and controlled-substance rules, and a plan for any licence or quota calendar (file the licence early in the year, with the fallback in section 9); any current chemical-control licence required locally for assay reagents; the rooms listed below; a dedicated reversed-cycle room, six arenas, a water maze, two Hebb-Williams mazes and a separate euthanasia room; written record of ICV cannulation, set-shifting and water-maze work; access to all listed equipment; references from two previous external collaborations; a named roster built to local labour-law limits (daily and weekly hours, overtime caps, a break on long shifts, a weekly rest, premiums for Sunday, holiday and night work, and a minimum rest between shifts; lights-off is set at about 07:00 so that the 13-h day runs 06:00–19:00 and avoids night hours; the roster covers days –15 to 37, about 53 consecutive days including weekends; each role states its employment regime and the rules that apply to it; facilities may close on holidays, so day 0 is scheduled to avoid public holidays and institutional recess; seven-day operation with weekly rest needs about 25–40% more people than the concurrency figures (about 15–21 named people for 12–15 at the busiest hours); student research stipends are a cheaper labour source to consider) of operators (with animal-research training records as required locally; about 12–15 people at the busiest hours, about 6

Item	Responsibility / requirement
	on ordinary days) and a facility-hours map for nights and weekends; stable operation (no recent strike disruption); a stated procedure for receiving foreign funds. Legal nature of the host (public body, public or private foundation, or private entity): some public hosts may import directly under their own exemptions, while a private host may have no such route and pay full import duties and taxes, so the import-duty scenario depends on the host type; a service contract may carry service taxes unless it qualifies as a service export (check locally); reassigning the host after Phase 0a is a restart (new ethics approval, fresh purchases, duty-free reagents cannot simply be transferred, about the Phase 0a cost again). Rooms and equipment: a dedicated reversed-cycle colony room; two ventilated set-shift rooms; a Hebb-Williams room; the water-maze room; six ELE/control arenas; light-tight or covered transport between rooms; infrared cameras and illuminators; a separate euthanasia room; the clock times of lights-off stated. Trade-off: a CRO offers fee-for-service and quality systems but rarely these assays; an academic lab offers expertise but expects authorship and slower timelines.
Payment tied to gates	Public hosts and foundations generally spend only deposited funds, so the first tranche must pre-fund Phase 0a: (1) at signature (ethics-committee fee, practice and satellite rats, the FGF17 pack and ERK antibodies, surgery, personnel for 0a); (2) after the written 0a gate decision (including the 14 placement scores), the main cohort; (3) at data lock, a small holdback. A failed 0a stops spending at its tranche; staff commitments on monthly stipends are stated for that case. The agreement states who bears currency risk: budget in the local currency, convert per tranche, and hold a 10–15% FX reserve (or agree USD tranches with a renegotiation band); bank spread and transfer taxes are included.
Governance and welfare oversight	Protocol amendments need the host PI, the sponsor and the ethics committee; the veterinary lead can stop the study; weekly remote welfare report (weights, scores, cap losses, adverse events); escalation: sponsor within 48 h, ethics-committee coordinator and unit head within 7 days; fixed weekly call slot.
Quality standard	Non-GLP; the host works to GLP-like good practice (contemporaneous records, calibrated equipment, training records, retained raw data).
Roles and accountability	Host PI (accountable for ethics-committee compliance and animal welfare; also the PI on the ethics application); study coordinator (one named person); about six operators on ordinary days and about twelve to fifteen at the busiest hours, in two staggered shifts as in section 5 (ELE staff, blinded injector, testers and preparer kept separate), two surgeons and a weekend rota; an unblinded technician outside the testing team who prepares coded syringes; sponsor (design, randomisation, drug-code key, registration, independent analysis).
Agreement terms	Funding, overhead and payment schedule; who pays for overruns and for a failed Phase 0a; IP ownership; data and tissue ownership; insurance and liability; stop rules; publication and authorship (ICMJE criteria); data sharing.
Host-competence evidence	Before signature: the lab's written records of ICV cannulation, set-shifting and water-maze work, its western blot CV from previous work and a verified equipment list. Before the main cohort: ≥80% correct cannula placement (≥12 of 14) across the 12 satellite and 2 practice rats, scored from the fresh coronal slab through the cannula track (if it fails, the sponsor funds a short training and re-test or reassigns the host, and no main-cohort money is spent until it is met). Sponsor decides go/no-go on this evidence.
SOP pack (host completes with photos; sponsor reviews)	Surgery and cannula care; ICV bolus injection; vehicle preparation and dosing; ELE and control sessions; set-shifting; Hebb-Williams; water maze incl. temperature/hypothermia rules; food restriction and weighing; euthanasia, tissue banking; satellite dissection and western protocol; per-day time-budget table; welfare scoresheet; data-recording SOP; Appendix A parameters signed off. The ethics-committee application is drafted jointly: the sponsor supplies a model narrative (severity, endpoints, justification), the host adapts it.
Blinding and randomisation	Sponsor generates the allocation and the coded vial/arm labels (an unblinded host technician outside the testing team prints and fills them), and keeps the key; host testing staff handle coded IDs only; unblinding after data lock.
Data integrity and monitoring	Bound or electronic lab notebooks with audit trail; raw files hashed on upload; protocol deviations classed minor/major with major ones reported within 48 h and a sponsor decision; scheduled live-video audits and source-data verification of ≈10% of records; weekly summary (enrolment, weights, exclusions, adverse events, cap losses); raw data and images shared with the sponsor for independent blinded analysis.
Tissue transfer	Banked tissue stays at the host unless a material transfer agreement and the necessary export permits are in place (export permit requirements to be confirmed).
Decisions	After 0a and 0b the sponsor issues a written decision (provisional go / stop / redesign) before further spending.

Appendix A. Task Parameters (Defaults; Host to Confirm and Sign Off Before Animals Arrive)

Task	Parameters
Set-shifting (digging; Birrell & Brown 2000; Tait et al. 2018 [11,19])	Two ceramic pots per trial; stimulus dimensions: digging medium and odour (neither is used in the ELE); 3 days digging habituation to a baited pot; example exemplar sets (host to fix ≥ 12 each before animals arrive, because simple, compound, ID1, ID2 and ED each need new pairs and the training pairs are separate; drop near-duplicate media such as sawdust and wood chips); odours — cinnamon, nutmeg, clove, cumin, thyme, vanilla, coriander, ginger, paprika, anise, cardamom, oregano; media — fine sand, coarse aquarium gravel, glass marbles, paper strips, coconut fibre, cork granules, pumice chips, rice hulls, shredded cardboard, terracotta chips, slate chips, wood wool (any material the host uses as bedding or nesting is excluded) (pairs are matched for salience in the practice rats; dimension order is counterbalanced 3:2 across the 5 rats per arm in each half and recorded); stages: simple discrimination, compound discrimination, first intradimensional shift (ID1), second intradimensional shift (ID2, to build the attentional set), extradimensional shift (reversal stages omitted to limit session length — declared); validity check: ED trials > ID2 trials cohort-wide; dimension order and exemplar sets counterbalanced across rats and arms; odour applied to the medium and bait buried in both pots (masked), with a fixed discovery-trial rule; criterion 6 consecutive correct; maximum 60 trials per pre-ED stage and N_cap at ED; 3-min trial limit; test cap 2.25 h per rat per day; error on first incorrect dig; bait $\approx \frac{1}{2}$ cereal piece; pot positions pseudo-random; training cap 5 days.
Hebb-Williams (Rabinovitch & Rosvold 1951 [22])	Closed-field maze with barrier layouts; 2 days adaptation with food reward including the standard practice problems; 4 test problems chosen from the standard 12 by host and sponsor before animals arrive (fixed order, same layouts for all rats; investigator-chosen subset, not validated); 6 trials per problem (3-min cutoff), one problem per day (4 days), two identical mazes in parallel; error zones defined in advance; split-half reliability reported; dim white light (≈ 10 –20 lux) in the maze room, as for the water maze.
Morris water maze	Pool ≈ 150 –180 cm, 25 ± 1 °C, hidden 10-cm platform 1.5–2 cm below the surface in opaque water (non-toxic black tempera paint, for contrast with albino rats), one pool, ≥ 4 distal cues, dim white light (≈ 10 –20 lux, measured; spectrum recorded) in the pool room only — a declared exception to the dark-phase red-light rule; rats travel in covered cages and the colony room stays dark; 4 trials/day, 60 s maximum, four quasi-random start points, ≥ 5 min between trials in a warm cage (32 °C); temperature screened with the implanted transponder at the end of each session, and a rest day if the confirming rectal reading is < 34.5 °C; 4 acquisition days; 60-s probe 24 h after the last acquisition day; visible-platform test (4 trials) the day after the probe (or the same day, ≥ 1 h later, for a rat whose acquisition was shifted by a rest day); scores: mean proximity (Gallagher), target-quadrant time, swim speed, floating, thigmotaxis.
ICV cannula (to be confirmed in the practice group)	Right lateral ventricle, rat aged 8–10 weeks (≈ 250 –300 g): from bregma AP -0.9 mm, ML $+1.5$ mm, DV such that the injector tip lies at -4.0 to -4.5 mm (the host states whether DV is measured from bregma or the skull surface; guide ≈ -3.0 to -3.5 mm with the injector protruding ≈ 1 mm), scaled by the bregma–lambda distance and confirmed by dye in the practice group; fixed with screws and dental cement under a protected recessed cap; dummy cannula between injections.
FGF17 bioassay (0a reagent gate)	Serum-starved FGF-responsive cells (the supplier’s BALB/3T3 line, or NIH-3T3; primary rat OPCs if available) stimulated for 10–15 min with 0–100 ng/ml of the supplied FGF17 in the presence of heparin 10 μ g/ml (an EC50 comparison is only valid with matching heparin); pERK/total ERK by western; FGF2 as positive control; pass = a clear dose-response with ≥ 3 -fold maximal pERK over unstimulated cells; the supplier’s stated ED50 is a proliferation value, so on first purchase the pERK dose-response is judged against FGF2 and the ≥ 3 -fold criterion (an earlier-lot comparison is only possible for later lots). A 3T3 pass confirms potency through FGFR1c/2c, not the FGFR3c/FGFR4 signalling that FGF17 prefers (FGFR3, with FGFR1, is the oligodendrocyte-precursor-relevant receptor). The Cohesion protein is E. coli-derived with a His tag. The boiled (heat-inactivated) aliquot is run in the same assay and must show $< 10\%$ residual pERK activity and no visible aggregation.
Antibodies and kits (host to confirm; unverified examples from memory)	pERK1/2 Thr202/Tyr204 (e.g. CST #4370), total ERK1/2 (e.g. CST #4695); the host lists final catalogue numbers, rat reactivity and dilutions in the SOP pack.
ELE / control session	Dark phase under red light; solo session of 60 min (no partner); 6 arenas (4 ELE, 2 control), arena identity rotated by day within type; arms balanced within every time block (two rats per arm per block); log object list, puzzle level, rewards, engagement minutes, wheel revolutions.

Appendix B. Operator Time Budget (to Be Confirmed by the Host)

Period	Daily work	Operator-hours/day
Days -15 to -13	Cannula surgery, ≤10 rats/day on two frames (two surgeons + two assistants, ≈8 h each)	≈ 32
Days -12 to -1	Handling and weighing ≈ 8 h/day; arena habituation from day -7 (30 rats in an empty arena) adds ≈ 12 h/day for 7 days	≈ 8 / 20
Days 0–34, every day incl. weekends (routine)	ELE/control: 30 solo sessions in 6 arenas over five blocks (≈6.5 h of sessions; one operator per two arenas ≈ 20 h, plus a fourth ELE helper for resets ≈ 4 h); injections ≈ 1.5 h; weighing, individual rations and cage checks ≈ 1.5 h; cleaning and object preparation ≈ 2 h; logs and video ≈ 1 h	≈ 30
Days 0, 2, 4, 6 (extra)	ICV boluses (30 rats × ≈10 min × 2 operators)	+ 10
Days 5–10	Food restriction ramp, extra weighing	+ 2
Days 11–16	Half A set-shift habituation/training (15 rats × ≈1 h); half B Hebb-Williams (15 rats × ≈0.5 h)	+ 22 (total ≈ 52)
Days 17–18	Half A training 15 h; half B set-shift habituation 15 h	+ 30 (total ≈ 60)
Days 19–22	Half A set-shift tests: up to 10 rats testing × ≤2.25 h ≈ 22 h (days 20–21 are the busiest; at most about 4 testing at one moment); half B habituation (day 19) or training (days 20–22) ≈ 15 h	+ 37 (total ≈ 67; ≈ 75 with maintenance and the technician)
Days 23–24	Half A Hebb-Williams ≈ 8 h; half B set-shift training ≈ 15 h	+ 23 (total ≈ 53)
Days 25–28	Half A Hebb-Williams ≈ 8 h; half B set-shift tests ≈ 22 h	+ 30 (total ≈ 60)
Days 29–34 (water maze)	30 rats × 4 trials, interleaved by block, two core temperatures ≈ 9 h with two operators	+ 18 (total ≈ 48)
Days 35–36 (euthanasia)	whole cage pairs (8 + 7 a day) with two teams, an extra tissue person per team, ≤12 min per animal; no routine (ELE and dosing end on day 34)	≈ 12
Weekends	The same task schedule repeats on weekends (day 0 on a Thursday)	as the weekday row
Phase 0a (before the cohort)	14 surgeries ≈ 35 h; tolerability dosing and palpation (35 days) ≈ 35 h; satellite dosing and dissection ≈ 15 h; bioassay, potency and westerns ≈ 80 h; training and set-up ≈ 85 h	≈ 250 in total
Maintenance and reminder sessions (half A on days 19–20, half B on days 25–26)	rats waiting to test hold on one maintenance session a day; day-before reminders (maintenance during the training windows is already in those rows)	+ 7 on 4 days (≈30 in total)
Pilot rats (Phase 0a, weeks -8 to -4)	habituation, restriction, training, ED-pace and salience runs, IPTT calibration swims, tester training	≈ 60 in total
Unblinded technician	daily preparation of coded syringes and vials, 35 days	≈ 35 in total
Video verification	Set-shift and Hebb-Williams sessions verified from video and 20% re-scored	≈ 150 in total

Total ≈ 2,550 operator-hours: the main cohort rows sum to about 2,020 (routine 35 × 30 = 1,050; extras 40 + 12 + 132 + 60 + 148 + 46 + 120 + 108 = 666; surgery 96; handling and habituation 180; euthanasia 24), plus maintenance ≈ 30, technician ≈ 35, pilot rats ≈ 60, Phase 0a ≈ 250 and video ≈ 150 (2,020 + 30 + 35 + 60 + 250 + 150 = 2,545). The peak is ≈ 75 operator-hours a day on set-shift test days (days 20–21; days 19 and 22 have only 5 rats testing and need less); the routine rows count 30 ELE/control sessions every day, about 1 h less on a rat's first test day. Because dosing is staggered over 5 hours and behaviour starts are staggered by block, about 4 testers and about 4 trainers are needed at once (about 8, rostered as about 10 to cover breaks); with ELE staff (4), the blinded injector and weigher (2) and the other half's operators, the busiest hours need about 12–15 people in two staggered shifts (75 operator-hours cannot be covered by 12–15 people on 8-h shifts alone: some are part-time or stipend holders on short blocks, or the roster is larger), and ordinary days about 6. Weekend and other labour-law premiums are part of the unquoted personnel cost. Personnel cost is not in the budget.

11. Validation Record

The mini design inherits its task methods, assay logic and many safeguards from a larger design reviewed in the rounds below (the early rounds were run in the earlier form). Reviewer panels were 5 Opus + 2 Sonnet agents per round unless stated. Reviewer statements are model judgements, not independent human review.

Round	Reviewers	Verdicts	Main issues → fix
Earlier A, B, translational	3 Opus, 3 Opus + 1 Haiku, 1 Opus	All approve with changes	Pump ends before testing, pump cost, middle-aged rats, transfer overlap, social confound, composite not a g factor, overclaiming, neurotrophin history, FGF safety → boluses, young adults, staged gates, hedged roadmap.
earlier-form rounds 1–8	5 Opus + 2 Sonnet each	All approve with changes every round	Gate calibration, slab fixation, vehicle choice, sequential waves, fixed slots, set-shift scoring, staffing arithmetic, frozen-SD rule, Cox primary, time-matched vehicle, heat-inactivated control, tolerability, budget chain, many wording fixes (history of the larger design).

Round	Reviewers	Verdicts	Main issues → fix
Full-size round B1	5 Opus + 2 Sonnet	All approve with changes; budget and executability: major revision	Housing and rats underpriced; all-in cost unquantified; wrong legal instruments; import quota exhausted in 2026; controlled-drug licences; dark-phase arithmetic on test days; weekend contradiction; staffing below concurrency; ill-defined gate and re-test → this mini redesign (smaller, one wave, solo ELE sessions, parallel halves, gate and OC re-simulated, budget re-itemised).
Mini round M1	5 Opus + 2 Sonnet	All approve with changes; behaviour: not executable as written (blocking, fixable on paper); budget: all-in cost understated	Daily block model could not carry the 25–30 pre-dose sessions → dosing ladder staggered 2.5–7.5 h after lights-off with fixed blocks and divider cages; day 0 on a Thursday; ≥10 set-shift exemplars and stage-failure/time-cap rules; budget re-itemised (base ≈6–9k → ≈9.5–13.5k: arenas and mazes, cannulas, two imports, small consumables); all-in 25–60k; tramadol and controlled-drug route, severity grading, 2027 service-tax regime, operator pay by contract route; decision-rule OC by effect location, attrition rule, gate sensitivity; EdU removed; many wording fixes.
Mini round M2	5 Opus + 2 Sonnet	All approve with changes; blocking: set-shift start-day allocation, time-limit and censoring rules, FX reserve vs ceiling, sponsor legal entity, route (a) vs (b)	Latin-rotation start days by block; protected ED budget, stage-failure and time-censoring rules (capped in the index, censored in Cox); home-cage interval on test day 1; fourth ELE operator (operator-hours 2,530; peak 75); satellite punch-first order; whole-pair euthanasia; analgesia regimen and humane endpoints; budget shown with FX reserve and 0a/0b itemised; tax-law update; import-tax attestation corrected; controlled-drug authorisation limits; pay by contract route; sponsor entity; host types; wording fixes.
Mini round M3	5 Opus + 2 Sonnet	All approve with changes; blocking: day-2 ID2 time rule vs stage-failure rule, 2027 tax/quota base case and controlled-drug clearance, set-shift worst-score handling	ID2/carry-over stop at 60 min, ED fixed 75 min, every non-completer scored at the cap (intention-to-test), Latin start-day table, test-day-1 home-cage interval anchored, tester rotation, MWM rest-day and IPTT rules, welfare endpoints, coded-photo necropsy; budget: re-test cost, distributor scenario, FX range, operator range 6–24k, reserve counted against the ceiling; regulation: controlled-drug authorisations, tax and import rules, 2027 regime, cell-line clearance, ethics-rule updates, FX and donation-tax citations, ICH S6; abbreviations list; stale text removed.
Mini round M4	5 Opus + 2 Sonnet	All approve with changes; blocking: trainer/box concurrency on test days, worst-score scope (deaths and cap loss), ED-pace pilot animals, small consumables and base budget, severity grade inconsistency	Cap scoring limited to behavioural reasons, training non-starters reported separately, time-out rule merged and made per-arm, ED budget up to 75 min after a 55-min ID2 stop, clock-time timetable, pilot data source, IPTT calibration, greyscale necropsy photos, bait accounting, distinct media; budget base ≈9.5–13.5k → ≈10–14k (transponders, consumables), reserve and ceiling 14–15k, re-test third import, all-in 25–60k, 2,530 operator-hours and peak 75; regulation: import licensing per shipment, host-type rows, route (b) 2027 taxes, controlled-drug and tax-law wording, FX classification, FX range; Fortin 2005 correction.
Mini round M5	5 Opus + 2 Sonnet	All approve with changes; no redesign needed; blocking-on-paper: set-shift scoring wording across sections, N_cap vs 60, pilot rats missing from totals, shipment count and Q1 purchase base case	One scoring definition (cap only for pre-ED failure, N_cap or time-out; non-starters separate; deaths, humane endpoints and cap loss missing), N_cap from the pilot frozen in OSF, 4 pilot rats in the totals (52, up to 64), set-shifting-alone always Cox and no data-driven trigger, Cox censoring rules, rooms split, arena turnover, inert media, half-B table, temperature rule; budget: four shipments (1.0–1.5k), transponders 0.5–0.75k, base 9–15k, materials budget up to about 15k, re-test 10–16k and 10–18 weeks, Q1 base-case purchase and 1.0–1.5k sunk cost, all-in 25–65k, +25–40% headcount; regulation wording, Fortin 2005. This was the last review round: this version 6 was not reviewed again.

12. Items the Reviewers Still Do Not Like / Remaining Gaps

#	Gap	Kind
1	Bolus FGF17 is not the Iram infusion ($\approx 7\times$ less volume, pulsatile); the dose approximately matches Iram's reported $\approx 2.1\ \mu\text{g}$ total mass, but that figure rests on one reviewer's retrieval and exposure time is not matched; a null remains confounded with delivery.	Pumps in the larger study
2	Power: only very large effects detectable ($d \approx 1.6\text{--}1.8$ at 6–7 per arm); healthy rats near ceiling; the direction rule is lenient ($\approx 10\%$ false-go; $\approx 39\%$ go for enrichment alone), is driven by the spatial tasks ($\approx 25\%$ go if the effect is on set-shifting only) and cannot separate the drug from the environment because there is no drug-only arm.	Intrinsic to a mini pilot
3	The FGF17 gate uses one time point (30 min) and 4 rats per group; a true 1.5-fold engagement passes only about half the time, the heat-inactivated control and an assay CV $\leq 50\%$ are assumed, and the time course is an assumption; periventricular pERK shows bioactivity at the ventricular interface only. The Iram 2022 dose anchor (and so the single-pack purchase) must be confirmed from the Methods before ordering. The heat-inactivated control can still produce non-specific pERK.	Intrinsic; assay needed
4	7,8-DHF may be inert in vivo; 35 injections at the long end of precedent; the tolerability screen (2 rats per vehicle) only catches gross intolerance; DHF is not gated.	Declared
5	The ELE is solo, non-social and a weak dose; set-shifting is the only clean transfer readout; the arm cannot be fully blinded; parallel halves use opposite task orders (order effects are a fixed effect, arm $\times$ half not estimable).	Declared
6	Prices unverified (local per-diem housing cost not found); the base materials cost ($\approx 9\text{--}15\text{k}$) can exceed a materials budget of about USD 15k at its upper end, and the pessimistic cases reach $\approx 12.5\text{--}18\text{k}$, $\approx 14\text{--}20\text{k}$ and (with a local distributor) $\approx 15\text{--}23\text{k}$; personnel, overhead, equipment fees and a currency reserve are excluded and the all-in cost is probably USD 25–65k (unverified; the host must quote); import exemptions, quotas and taxes vary by country and are uncertain (import duties and fees are not included; losing an import exemption adds $\approx 1.2\text{--}2.2\text{k}$ and months).	Quotes and written confirmations
7	Host executability depends on a named roster (about 12–15 people at the busiest hours, about 6 on ordinary days), rooms and equipment, ethics-committee and animal-facility licence status, a controlled-drug route under national controlled-substance rules, a legal entity that can contract for the individual sponsor, a legal instrument and a pre-funded first tranche, none of which are in hand.	Host to supply
8	Structure (Olig2/MBP histology) and synaptic readouts are not analysed in this study; tissue is only banked.	Larger study
9	Optogenetics (Stage 1b) and the human roadmap are aspirations without design detail or validated tools in the host site.	Own review round
10	Literature checks used search snippets only; Iram's delivered dose was reported by one reviewer only; the FGF1 ICV pERK precedent is not referenced; human-trial and neurotrophin statements to be verified; local regulatory, import and price details were not verified and must be confirmed by the host.	Researcher/host to verify
11	The tolerability screen (2 rats per vehicle) cannot compare vehicles; the vehicle is a 35-day IP exposure in all arms; the periventricular gate uses one time point and 4 rats per group; the analysis model is ambitious for $\approx 18\text{--}21$ completers (a simpler fallback is pre-registered).	Declared

13. References and Verification Status

Checks were by search snippets only (PubMed, PMC, Crossref and publisher pages unreachable). “Verified” means consistent across indexed records.

#	Citation	Status
1	Iram T, Kern F, Kaur A, et al. Young CSF restores oligodendrogenesis and memory in aged mice via Fgf17. Nature 2022;605:509–515. doi:10.1038/s41586-022-04722-0 (an Author Correction exists)	Verified; delivered dose (25 $\mu\text{g}/\text{ml}$; 0.5 $\mu\text{l}/\text{h} \times 7\text{ d} \approx 2.1\ \mu\text{g}$) reported by one reviewer, not independently confirmed
2	Boltaev U, et al. Multiplex quantitative assays indicate a need for reevaluating reported small-molecule TrkB agonists. Sci Signal 2017;10(493). doi:10.1126/scisignal.aal1670	Verified
3	Todd D, Gowers I, Dowler SJ, et al. A monoclonal antibody TrkB receptor agonist as a potential therapeutic for Huntington's disease PLoS One 2014;9(2):e87923. doi:10.1371/journal.pone.0087923	Citation verified; DHF content via reviews
4	Pankiewicz P, et al. Do small molecules activate the TrkB receptor in the same manner as BDNF? Limitations of published TrkB low molecular agonists and screening for novel TrkB orthosteric agonists. Pharmaceuticals 2021;14(8):704. doi:10.3390/ph14080704	Verified (review)
5	Rawlings-Mortimer F, et al. 7,8-dihydroxyflavone enhances long-term spatial memory and alters brain volume in wildtype mice. Front Syst Neurosci 2023;17:1134594. doi:10.3389/fnsys.2023.1134594	Verified
6	van Praag H, Kempermann G, Gage FH. Nat Rev Neurosci 2000;1:191–198. doi:10.1038/35044558	Verified
7	Nithianantharajah J, Hannan AJ. Nat Rev Neurosci 2006;7:697–709. doi:10.1038/nrn1970	Verified
8	Galsworthy MJ, et al. Behav Genet 2005;35:675–692. doi:10.1007/s10519-005-3423-9	Verified

#	Citation	Status
9	Matzel LD, Han YR, Grossman H, et al. J Neurosci 2003;23(16):6423–6433 (general learning ability in mice). doi:10.1523/JNEUROSCI.23-16-06423.2003	Verified
10	Gibson EM, Purger D, Mount CW, et al. Neuronal activity promotes oligodendrogenesis and adaptive myelination in the mammalian brain. Science 2014;344:1252304 (mouse optogenetic premotor-cortex stimulation)	Verified
11	Birrell JM, Brown VJ. Medial frontal cortex mediates perceptual attentional set shifting in the rat. J Neurosci 2000;20(11):4320–4324 (published protocols differ in stages; some add ID2, which is used here as a modification)	Verified
12	Kraemer HC, Mintz J, Noda A, Tinklenberg J, Yesavage JA. Caution regarding the use of pilot studies to guide power calculations for study proposals. Arch Gen Psychiatry 2006;63:484–489. doi:10.1001/archpsyc.63.5.484	Verified
13	Leon AC, Davis LL, Kraemer HC. The role and interpretation of pilot studies in clinical research. J Psychiatr Res 2011;45:626–629. doi:10.1016/j.jpsychires.2010.10.008	Verified
14	Percie du Sert N, et al. PLoS Biol 2020;18:e3000410. doi:10.1371/journal.pbio.3000410	Verified
15	Enrichment meta-analysis (rodent cognition $g \approx 0.44$, 95% CI 0.25–0.62): Fitzgerald, Univ. at Albany dissertation, a cross-species meta-analysis of environmental enrichment effects on cognition (scholarsarchive.library.albany.edu/etd/371); Léger M, et al. Environmental enrichment duration differentially affects behavior and neuroplasticity in adult mice. Cereb Cortex 2015;25(11):4048–4061. doi:10.1093/cercor/bhu119	Dissertation is not peer-reviewed; Léger verified
16	Fortin D, Rom E, Sun H, Yayon A, Bansal R. Distinct FGF/FGF receptor signaling pairs initiate diverse cellular responses in the oligodendrocyte lineage. J Neurosci 2005;25:7470–7479. doi:10.1523/JNEUROSCI.2120-05.2005	Verified
17	Nie S, et al. 7,8-DHF protects nigrostriatal dopaminergic neurons from rotenone-induced neurotoxicity in rodents. Rotenone model in 9-month-old rats, 7,8-DHF 5 mg/kg IP daily for 5 weeks. Parkinsons Dis 2019;2019:9193534. doi:10.1155/2019/9193534 (PMC6421741)	Verified
18	Gallego-Ortega A, et al. 7,8-Dihydroxiflavone maintains retinal functionality and protects various types of RGCs in adult rats with optic nerve transection. Int J Mol Sci 2021;22(21):11815. doi:10.3390/ijms222111815 (5 mg/kg IP daily; outcomes at 7, 14 and 21 days; the 1% DMSO detail was not confirmed)	Verified
19	Tait DS, Bowman EM, Neuwirth LS, Brown VJ. Assessment of intradimensional/extradimensional attentional set-shifting in rats. Neurosci Biobehav Rev 2018;89:72–84	Verified
20	Browne RH. On the use of a pilot sample for sample size determination. Stat Med 1995;14:1933–1940	Verified (title to check)
21	Gallagher M, Burwell R, Burchinal M. Severity of spatial learning impairment in aging: development of a learning index for performance in the Morris water maze. Behav Neurosci 1993;107:618–626. doi:10.1037/0735-7044.107.4.618	Verified
22	Rabinovitch MS, Rosvold HE. A closed field intelligence test for rats. Can J Psychol 1951;5:122–128 (Hebb-Williams maze)	Verified (issue number unconfirmed)
23	Zhang X, et al. Receptor specificity of the fibroblast growth factor family: the complete mammalian FGF family. J Biol Chem 2006;281:15694–15700 (FGF8 subfamily expected to act through FGFR3c/FGFR4; FGF17-specific profile not retrieved)	Real; FGF17 result unverified

Not found: any study of FGF17/FGF-family treatment improving learning in healthy young rodents; any study combining 7,8-DHF with enrichment; rat ICV FGF17 doses; independent confirmation of Iram 2022’s delivered dose; local prices for Alzet pumps and recombinant FGF17; a peer-reviewed validation of a Hebb-Williams subset; a local per-diem housing cost. Local regulatory, import and price details were not verified; the host must confirm them.